

418-HW1

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EXERCISE 1

2i

```
A = matrix(c(1,1,-2,1,0,1,-2,1,-1),3,3,byrow=TRUE)
decomposition = eigen(A)
cat('Eigenvalues of A: \n')
```

```
## Eigenvalues of A:
```

```
decomposition$values
```

```
## [1]  2.2924016  0.6027049 -2.8951065
```

```
cat('Eigenvectors of A: \n')
```

```
## Eigenvectors of A:
```

```
decomposition$vectors
```

```
##           [,1]      [,2]      [,3]
## [1,]  0.8642795 0.07589338  0.4972536
## [2,]  0.1705987 0.88573564 -0.4317041
## [3,] -0.4731987 0.45794385  0.7525758
```

2ii

```
A = matrix(c(1,1,-2,1,0,1,-2,1,-1),3,3,byrow=TRUE)
n = 3 # dimension of matrix
trace = 0 # initialize trace as 0 before summing
for (i in 1:n) { # for loop to add each diagonal entry
  trace = A[i,i] + trace # diagonal entry: row index = column index
}
cat('Trace of A: \n')
```

```
## Trace of A:
```

```
trace
```

```
## [1] 0
```

```
cat('Determinant of A: \n')
```

```
## Determinant of A:
```

```
det(A)
```

```
## [1] -4
```

```
decomposition = eigen(A)
```

```
values = decomposition$values
```

```
cat('Sum of Eigenvalues of A: \n') # equivalent to trace(A) = 0
```

```
## Sum of Eigenvalues of A:
```

```
sum(values) # use sum function, extremely small result approx. equal to 0
```

```
## [1] 1.776357e-15
```

```
cat('Product of Eigenvalues of A: \n') # equivalent to det(A) = -4
```

```
## Product of Eigenvalues of A:
```

```
values[1]*values[2]*values[3]
```

```
## [1] -4
```

EXERCISE 3

3i

```
A = matrix(c(3,6,-1,6,9,4,-1,4,3),3,3,byrow=TRUE)
```

```
eigen = eigen(A) # eigendecomposition
```

```
eigenvalues1 = eigen$values # save components
```

```
lambda1 = eigenvalues1[1]
```

```
lambda2 = eigenvalues1[2]
```

```
lambda3 = eigenvalues1[3]
```

```
vector1 = eigen$vectors[,1]
```

```
vector2 = eigen$vectors[,2]
```

```
vector3 = eigen$vectors[,3]
```

```
spec = lambda1*matrix(vector1)*vector1 + lambda2*matrix(vector2)*vector2 +  
lambda3*matrix(vector3)*vector3 # spectral decomposition
```

```
cat('Spectral decomposition: \n')
```

```
## Spectral decomposition:
```

```
spec
```

```
##      [,1]
## [1,]    3
## [2,]    9
## [3,]    3
```

3ii

```
A = matrix(c(3,6,-1,6,9,4,-1,4,3),3,3,byrow=TRUE)
eigen = eigen(A)
eigenvalues = eigen$values
Asq = A %*% A # square matrix
sq_eigen = eigen(Asq)
sq_eigenvalues = sq_eigen$values
lambda1 = sq_eigenvalues[1]
lambda2 = sq_eigenvalues[2]
lambda3 = sq_eigenvalues[3]
vector1 = sq_eigen$vectors[,1]
vector2 = sq_eigen$vectors[,2]
vector3 = sq_eigen$vectors[,3]
spec = lambda1*matrix(vector1)*vector1 + lambda2*matrix(vector2)*vector2 +
       lambda3*matrix(vector3)*vector3
cat('Spectral decomposition: \n')
```

```
## Spectral decomposition:
```

```
spec
```

```
##      [,1]
## [1,]   46
## [2,]  133
## [3,]   26
```

```
D = diag(eigenvalues)
cat('Squared Eigenvalue Matrix for original A: \n')
```

```
## Squared Eigenvalue Matrix for original A:
```

```
D %*% D
```

```
##      [,1] [,2] [,3]
## [1,] 183.3781 0.0000 0.000000
## [2,]  0.0000 15.4864 0.000000
## [3,]  0.0000  0.0000 6.135498
```

```
cat('Eigenvalue Matrix for A^2: \n')
```

```
## Eigenvalue Matrix for A^2:
```

```
diag(sq_eigenvalues) # equivalent to A^2 eigenvalue matrix
```

```
##           [,1]      [,2]      [,3]
## [1,] 183.3781  0.0000  0.000000
## [2,]  0.0000 15.4864  0.000000
## [3,]  0.0000  0.0000  6.135498
```

3iii

```
A = matrix(c(3,6,-1,6,9,4,-1,4,3),3,3,byrow=TRUE)
eigen = eigen(A)
eigenvalues = eigen$values
Ainv = solve(A) # calculate inverse of matrix
inv_eigen = eigen(Ainv)
inv_eigenvalues = inv_eigen$values
lambda1 = inv_eigenvalues[1]
lambda2 = inv_eigenvalues[2]
lambda3 = inv_eigenvalues[3]
vector1 = inv_eigen$vectors[,1]
vector2 = inv_eigen$vectors[,2]
vector3 = inv_eigen$vectors[,3]
spec = lambda1*matrix(vector1)*vector1 + lambda2*matrix(vector2)*vector2 +
       lambda3*matrix(vector3)*vector3
cat('Spectral decomposition: \n')
```

```
## Spectral decomposition:
```

```
spec
```

```
##           [,1]
## [1,] -0.08333333
## [2,] -0.06060606
## [3,]  0.06818182
```

```
D = diag(eigenvalues)
cat('Inverse of Original A Eigenvalue Matrix: \n')
```

```
## Inverse of Original A Eigenvalue Matrix:
```

```
solve(D)
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.07384588  0.0000000  0.0000000
## [2,] 0.00000000  0.2541117  0.0000000
## [3,] 0.00000000  0.0000000 -0.4037152
```

```
cat('Eigenvalue Matrix of A-1: \n')
```

```
## Eigenvalue Matrix of A-1:
```

```
diag(inv_eigenvalues)
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.2541117 0.00000000 0.0000000
## [2,] 0.0000000 0.07384588 0.0000000
## [3,] 0.0000000 0.00000000 -0.4037152
```

```
# same eigenvalues, being displayed in a different order.
# likely due to order of operations within R. Program is not switching 0.254
# with 0.0738, the solve() function simply divides each of the original
# eigenvalues from 1 (e.g. 1/13.54 = 0.738). Since 13.54 is the largest original
# eigenvalue of A, its result remains in the first entry of the matrix.
```

4

```
# One of the main points of the section is that multivariate data sets are
# extremely common in practice, having spread to a plethora of disciplines and
# fields of study. Soon after, the author notes that the primary goal of
# multivariate data analysis is the simplification of data and inter-variable
# relationships. The opener also makes note of the advantage that multivariate
# data analysis does not require each variable to be on a similar scale (even
# performing well with ordinal data). It goes on to state that the analysis is
# predominantly concerned with descriptive and inferential statistics. The
# purpose of descriptive statistics being to determine the significance of
# variables and their respective contributions, with multivariate inferences
# being a sort of extension of univariate analysis. Lastly, the chapter places
# heavy emphasis on the fact that multivariate analysis is easily carried out
# with the development of computer softwares, dramatically expanding the scope
# of statistics in general.
```